Unit 2

Lesson Three

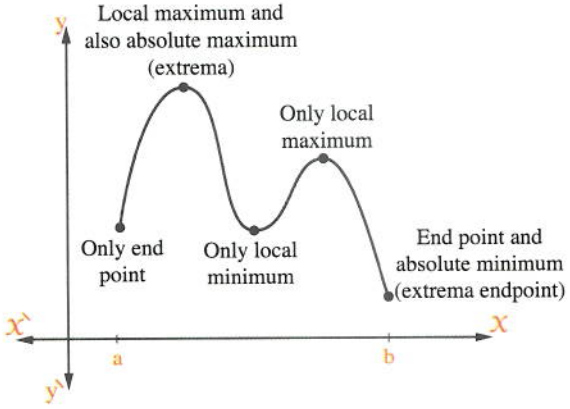
Absolute maxima and minima (Absolute extrema)

Definition

If the function f is defined on the closed interval $[a, b]$ and $c \in [a, b]$, then

1. $f(c)$ is an absolute minimum value on the interval $[a, b]$ when $f(c) \leq f(x)$ for each $x \in [a, b]$
2. $f(c)$ is an absolute maximum value on the interval $[a, b]$ when $f(c) \geq f(x)$ for each $x \in [a, b]$

- The absolute maximum value of a function on a closed interval is the greatest value in the range of the function f and the absolute minimum value is the smallest value in the range of the function f and these values are called the extrema of the function.
- The maximum value can be occured at an interior point of the interval or at the boundries of the function once it occurs at the boundries of the interval, it is called the end point extrema.

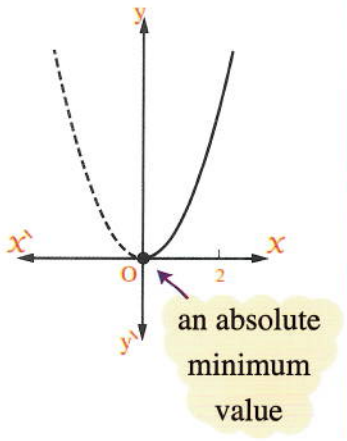
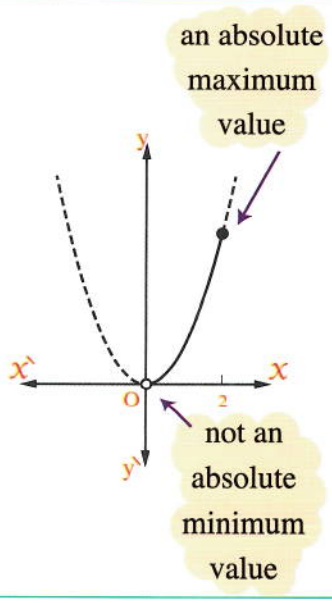
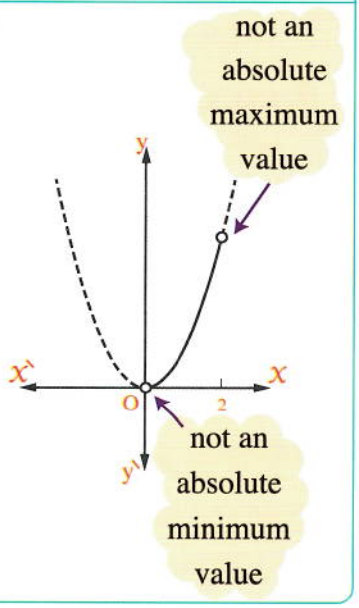


- The local maximum (or minimum) value of a function is the maximum (or minimum) value of this function in a neighbourhood of the point.

  i.e. Small part of the interval of a function is the maximum (or minimum) value that f is defined while the absolute maximum (or minimum) value is the greatest (smallest) value of this function in all the interval that f is defined.

- The absolute maximum value of a continuous function over the interval $[a, b]$ must be either a point of the local maximum values or one of the endpoints of this interval where it is the greatest value of them.
- The absolute minimum value of a continuous function over the interval $[a, b]$ must be either a point of the local minimum values or one of the endpoints of this interval where it is the smallest value of them.

⊙ The following figures illustrate the absolute maximum and minimum values of the function $f : f(x) = x^2$ according to its given domain.

the domain = $[0 , \infty[$	the domain = $]0 , 2]$	the domain = $]0 , 2[$
		

Theorem

If function f is continuous on the interval $[a , b]$, then the function f has an absolute maximum value and absolute minimum value on the interval $[a , b]$

Studying the absolute maximum and minimum values on a closed interval [a , b]

⊙ **If f is a continuous function on the interval $[a , b]$:**

1. Determine the critical points at which $f'(x)$ = zero or not exist and belongs to the interval $[a , b]$
2. Find the values of the function at the critical points and the endpoints $f(a)$ and $f(b)$
3. Compare among the previous values , then the greatest value is the absolute maximum value on $[a , b]$ and the smallest value is the absolute minimum value on $[a , b]$

Example 1

Find the absolute extrema values of the function $f : f(x) = 2x^4 - x^2 + 1$ on the interval $[-1 , 1]$

Solution

1. Find the values of x at which critical points ($f'(x) = 0$ or not exist)

$\because f(x) = 2x^4 - x^2 + 1$

$\therefore f'(x) = 8x^3 - 2x = 2x(4x^2 - 1) = 2x(2x - 1)(2x + 1)$

$\therefore f'(x) = 0$, when $x = 0$ or $x = \frac{1}{2}$ or $x = -\frac{1}{2}$

, $\because$ all the points $\in [-1, 1]$

$\therefore$ The critical points at $x = 0$, $\frac{1}{2}$, $-\frac{1}{2}$

2. Calculate the values of the function at the critical points which belong to the interval and at endpoints of $[-1, 1]$

$f(-1) = 2 - 1 + 1 = 2$, $f(1) = 2 - 1 + 1 = 2$

$f\left(-\frac{1}{2}\right) = \frac{1}{8} - \frac{1}{4} + 1 = \frac{7}{8}$, $f\left(\frac{1}{2}\right) = \frac{1}{8} - \frac{1}{4} + 1 = \frac{7}{8}$, $f(0) = 1$

$\therefore$ The absolute maximum value = 2 at $x = 1$ or -1

, the absolute minimum value = $\frac{7}{8}$ at $x = \frac{1}{2}$ or $-\frac{1}{2}$

Example 2

Find the absolute maximum and minimum value of the function

$f : f(x) = 1 + 12x - x^3$ on the interval $[1, 3]$

Solution

$\because f(x) = 1 + 12x - x^3$

$\therefore f'(x) = 12 - 3x^2 = 3(4 - x^2) = 3(2 - x)(2 + x)$

$\therefore f'(x) = 0$ when $x = 2$, $x = -2$

$\therefore$ The critical points at $x = 2$ or $x = -2$

, $\because x = -2 \notin [1, 3]$

The only critical point inside the given interval is at $x = 2$ and we find the values of the function at this point and at endpoints of the interval.

$\because$ The absolute maximum value = 17 at $x = 2$

, the absolute minimum value = 10 at $x = 3$

Example 3

Find the absolute extrema values of the function $f : f(x) = \frac{x}{x - 1}$ on the interval $[2, 4]$

Solution

$\because f(x) = \frac{x}{x - 1}$, $x \neq 1$

$\therefore f'(x) = \frac{(x - 1) - x}{(x - 1)^2} = \frac{-1}{(x - 1)^2}$

, $\because f'(x) < 0$ and defined for all values $x \in [2, 4]$

$\therefore$ There is no critical points of this function therefore we find only the values of the function at the endpoints of the interval.

$\therefore f(2) = \frac{2}{2-1} = 2$, $f(4) = \frac{4}{4-1} = \frac{4}{3}$

$\therefore$ The absolute maximum value $= 2$ at $x = 2$

, $\because$ the absolute minimum value $= 1\frac{1}{3}$ at $x = 4$

Example 4

Find the absolute maximum and minimum value of the function

$f : f(x) = \frac{\sqrt[3]{x-2}}{x}$ on the interval $[1 , 10]$

Solution

$\because f(x) = \frac{(x-2)^{\frac{1}{3}}}{x}$

$\therefore \grave{f}(x) = \frac{x \times \frac{1}{3}(x-2)^{-\frac{2}{3}} - (x-2)^{\frac{1}{3}} \times 1}{x^2}$

$\therefore \grave{f}(x) = \frac{\frac{1}{3}x(x-2)^{-\frac{2}{3}} - (x-2)^{\frac{1}{3}}}{x^2}$

$= \frac{\frac{1}{3}(x-2)^{-\frac{2}{3}}[x - 3(x-2)]}{x^2} = \frac{x - 3x + 6}{3x^2(x-2)^{\frac{2}{3}}} = \frac{6 - 2x}{3x^2(x-2)^{\frac{2}{3}}}$

$\therefore \grave{f}(x) = 0$ when $6 - 2x = 0$ **i.e.** When $x = 3$

$\therefore x = 3 \in [1 , 10]$ at which a critical point.

, $\because \grave{f}(x)$ does not exist when $x = 0$ or $x = 2$

, $f(x)$ is undefined at $x = 0$

$\therefore x = 2 \in [1 , 10]$ at which a critical point

, then we study the values of the function at $x = 3$, $x = 2$ and at the endpoints of the interval $x = 1$, $x = 10$

$\therefore f(3) = \frac{\sqrt[3]{3-2}}{3} = \frac{1}{3}$, $f(2) = \frac{\sqrt[3]{2-2}}{2} =$ zero

, $f(1) = \frac{\sqrt[3]{1-2}}{1} = -1$, $f(10) = \frac{\sqrt[3]{10-2}}{10} = \frac{2}{10} = \frac{1}{5}$

$\therefore$ The absolute maximum value $= \frac{1}{3}$ at $x = 3$

, the absolute minimum value $= -1$ at $x = 1$

Example 5

Find the absolute maximum and minimum values of the function

$f : f(x) = \begin{cases} x^3 + 3x^2 & , x \leq 0 \\ x^2 - 2x & , x > 0 \end{cases}$ **on the interval** $[-3, 3]$

Solution

• Investigate the derivative at $x = 0$

$$f'(0^+) = \lim_{h \to 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0^+} \frac{(0+h)^2 - 2(0+h) - 0}{h}$$

$$= \lim_{h \to 0^+} (h - 2) = -2$$

, $f'(0^-) = \lim_{h \to 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0^-} \frac{(0+h)^3 + 3(0+h)^2 - 0}{h}$

$$= \lim_{h \to 0^-} (h^2 + 3h) = \text{zero}$$

$\therefore f'(0)$ is not exist.

$\therefore$ There is a critical point at $x = 0$

$$\therefore f'(x) = \begin{cases} 3x^2 + 6x & , x < 0 \\ \text{not exist} & , x = 0 \\ 2x - 2 & , x > 0 \end{cases}$$

, at $x < 0$ $\quad \therefore f'(x) = 0$ when $3x^2 + 6x = 0$

$\therefore 3x(x + 2) = 0$ $\quad \therefore x = 0$ (refused) or $x = -2$

, at $x > 0$ $\quad \therefore 2x - 2 = 0$ $\quad \therefore x = 1$

$\therefore$ The critical point of the function is at $x = -2$ or 0 or 1

, then we find the values of the function at these points and also at the endpoints of the interval.

$\therefore f(-2) = (-2)^3 + 3(-2)^2 = 4$

, $f(0) = 0$ $\quad$, $f(1) = (1)^2 - 2(1) = -1$

, $f(-3) = (-3)^3 + 3(-3)^2 = 0$ $\quad$, $f(3) = (3)^2 - 2 \times 3 = 3$

$\therefore$ The absolute maximum value = 4 at $x = -2$

, the absolute minimum value = -1 at $x = 1$

Example 6

Find the absolute extrema values of the function

$f : f(X) = X^2 e^X$ on the interval $[-3 , 1]$

Solution

$\because f(X) = X^2 e^X$

$\therefore \grave{f}(X) = X^2 e^X + e^X \times 2X = X e^X (X + 2)$

, $\because \grave{f}(X) = 0$ when $X = 0$ or $X = -2$

, $\because \{0 , -2\} \subset [-3 , 1]$

$\therefore$ There is a critical points at $X = 0 , X = -2$

* Calculate the values of the function at the critical points and at endpoints of the interval $[-3 , 1]$

, $\because f(0) =$ zero , $f(-2) = (-2)^2 e^{-2} \simeq 0.54$

, $f(-3) = 0.45$, $f(1) = e \simeq 2.7$

$\therefore$ The absolute maximum value = e at $X = 1$

, the absolute minimum value = 0 at $X = 0$

Example 7

Find the absolute maximum and minimum value of the function $f : f(X) = \sqrt{25 - X^2}$

Solution

$\because f(X) = \sqrt{25 - x^2}$

$\therefore$ The domain of the function = $[-5 , 5]$

, $\because \grave{f}(X) = \dfrac{-2X}{2\sqrt{25 - x^2}} = \dfrac{-X}{\sqrt{25 - x^2}}$

Put $\grave{f}(X) = 0$ $\therefore X = 0$

, $\grave{f}(X)$ does not exist at $X = -5 , X = 5$ (endpoints)

$\therefore$ There is a critical points at $X = 0$ and there are two critical terminal points at $X = 5 , X = -5$

* Calculate the values of the function at the critical points and at endpoints of the interval $[-5 , 5]$

Notice that

There is a critical terminal points at $X = 5 , X = -5$ and at which no local maximum or minimum values but they are absolute minimum value.

$\therefore f(0) = 5$, $f(-5) = 0$, $f(5) = 0$

$\therefore$ The absolute maximum value = 5 at $x = 0$

, the absolute minimum value = 0 at $x = \pm 5$

Remark

If the function f is defined on the interval $[a , b]$ and if :

1. $f'(x) > 0$ **i.e.** The function is increasing on the same interval , then :

* The absolute minimum value = $f(a)$
* The absolute maximum value = $f(b)$

2. $f'(x) < 0$ **i.e.** The function is decreasing on the same interval , then :

* The absolute minimum value = $f(b)$
* The absolute maximum value = $f(a)$